

Topic 1.3 - Rates of Change in Linear and Quadratic Functions

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

Linear functions have constant rate. Quadratic functions do not, but their rates change in a constant pattern over equal input steps. This distinction supports model identification, construction, prediction, and comparison without relying only on graph appearance.

Learning targets

- Use first and second differences correctly, including the equal-spacing requirement.
- Construct linear or quadratic models from values, rates, and structural conditions.
- Derive and interpret how average rates of a quadratic change across equal intervals.

Language and structure

Term	Meaning in this topic
first difference	The change between consecutive outputs for equal input steps.
second difference	The change between consecutive first differences.
linear signature	Constant first differences for equally spaced inputs.
quadratic signature	Constant nonzero second differences for equally spaced inputs.
step size	The common input increment h used in finite differences.

Launch - reason before calculating

For $x = 0, 2, 4, 6$, outputs are 5, 13, 29, 53. A student says the data are not quadratic because first differences are not constant. Evaluate the claim.

Concept 1: Finite-difference signatures

Core idea

For equally spaced inputs, a nonconstant function with constant first differences is linear; constant nonzero second differences indicate a quadratic. If spacing is unequal, compare average rates instead of raw differences.

Worked example - annotate the reasoning

Problem. For $x = 0, 1, 2, 3, 4$, outputs are 3, 10, 21, 36, 55. Classify and find a rule.

Model reasoning. First differences 7, 11, 15, 19 have second difference 4, so $a = 2$. Since $f(0) = 3$ and $f(1) = 10$, $b = 5$. Thus $f = 2x^2 + 5x + 3$.

Check your understanding

Independent

No calculator

For outputs 1, 6, 17, 34, 57 at $x = 0, 1, 2, 3, 4$, find a rule.

Concept 2: Constructing a quadratic from values

Core idea

Three noncollinear points with distinct inputs determine a quadratic. When one input is 0, the constant term is immediate. Finite differences reveal the leading coefficient quickly; substitution then gives the remaining term.

Worked example - annotate the reasoning

Problem. A quadratic has $q(0) = 4$, $q(1) = 9$, $q(2) = 18$. Determine $q(x)$.

Model reasoning. First differences 5, 9 give second difference 4, so $a = 2$. Then $c = 4$ and $2 + b + 4 = 9$, so $b = 3$. Hence $q = 2x^2 + 3x + 4$.

Check your understanding

Independent

No calculator

Find the quadratic through $(0, 2)$, $(1, 7)$, $(2, 16)$.

Concept 3: How quadratic rates change

Core idea

For $q(x) = ax^2 + bx + c$, the average rate on $[x, x + h]$ is $a(2x + h) + b$. For fixed h , this is linear in the starting input. Consecutive rates separated by step h differ by $2ah$, and output second differences equal $2ah^2$.

Worked example - annotate the reasoning

Problem. For $f(x) = 2x^2 - 3x + 1$, compare rates on $[0, 2]$ and $[2, 4]$.

Model reasoning. Using $a(2x + h) + b$, rates are 1 and 9. They differ by $8 = 2ah$ when the starting input moves by 2.

Check your understanding

Independent

No calculator

For $q(x) = -x^2 + 6x$, find rates on $[0, 1]$, $[1, 2]$, $[2, 3]$.

Representation laboratory

Connect at least two representations

A table has inputs 0, 1, 3, 6 and outputs 2, 5, 17, 62. Explain why raw differences cannot prove a quadratic pattern. Compute consecutive average rates.

Challenge laboratory

Challenge - justify every claim

For $f(x) = ax^2 + bx + c$, prove $f(x+h) - 2f(x) + f(x-h) = 2ah^2$. Explain why b, c disappear.

Error clinic and calculator strategy

Common errors to diagnose

- Calling any changing first-difference table quadratic without checking second differences.
- Using raw differences when input increments are unequal.
- Assuming a quadratic pattern must continue outside the data domain.
- Equating a with the second difference when step size is not 1.

Technology decision

Regression helps with noisy data, but exact finite differences reveal structure more transparently. Inspect residuals, report spacing, and do not treat a high fit statistic as permission for unlimited extrapolation.

AP-style synthesis

Multi-part reasoning task

For equally spaced inputs, P has values

x	0	1	2	3	4
$P(x)$	12	18	28	42	60

A. Justify a quadratic model using differences.

B. Determine $P(x)$.

C. Find the average rate on $[1, 4]$ and connect it to one-unit rates.

Exit ticket

Before you leave

1. What spacing condition is required for raw finite differences?

2. For step 1, how is second difference related to a ?

3. Can a genuine quadratic have constant rate on all intervals?
